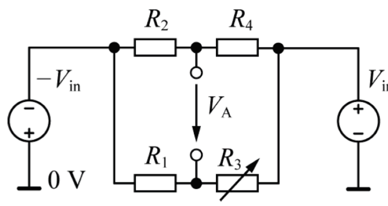
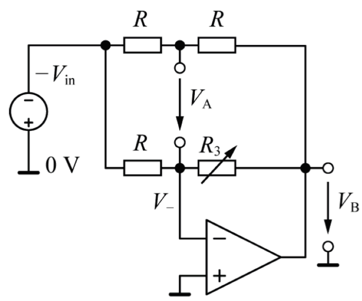


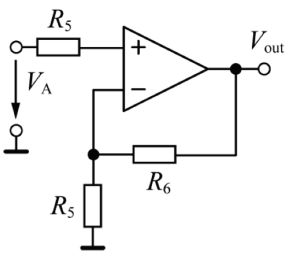

The circuit shown is a Wheatstone bridge that measures an unknown variable electrical resistance by balancing its two branches – one with the unknown component.

1. Find the expressions for the Thevenin voltage $V_{th}$ **[1p]** and Thevenin resistance $R_{th}$ **[1p]** of the circuit so that variable resistor $R_3$ appear in the numerator. The two terminals for this problem are the points where bridge voltage $V_A$ is measured. Express $V_A$ in terms of $V_{th}$, $R_{th}$ and the measuring resistance $R_m$ that would be present across the terminals if a voltmeter were connected. **[0.5p]**
2. The bridge is balanced when $R_1 = R_2 = R_3 = R_4 = R$. Simplify the expression for the bridge voltage as a function of the variable resistor $R_3 = R + \Delta r$ and the measuring resistance $R_m$, that is $V_A(\Delta r; R; R_m)$. **[1p]** If $V_{in} = 5$ V, $R = 2$ k$\Omega$ and $R_m = 100$ k$\Omega$, calculate the $V_A$ values for $\Delta r = \{-1 \text{ k}\Omega; 0 \text{ k}\Omega;\ +1 \text{ k}\Omega\}$. **[1p]**




Now, we remove voltage source $+V_{in}$ from the circuit and drive the bridge with an ideal op-amp as shown.

3. What is the potential $V_-$ of the inverting node and why? **[0.5p]** Express the op-amp voltage $V_B$ in terms of $R$, $R_3$ and $V_{in}$ when nothing is connected to read $V_A$ and explain why this expression is simple! **[1p]** Express the bridge voltage as a function of the variable resistor $R_3 = R + \Delta r$ and $V_{in}$, that is $V_A(\Delta r; R; V_{in})$. **[1p]**
4. Calculate $V_A(\Delta r)$ again for $V_{in} = 5$ V, $R = 2$ k$\Omega$ and $\Delta r = \{-1 \text{ k}\Omega; 0 \text{ k}\Omega;\ +1 \text{ k}\Omega\}$. **[0.5p]** Comparing the values to those in Problem-2, what are the advantages of this circuit in terms of $V_A(\Delta r)$ linearity and symmetry? **[1p]**




To take advantages of the circuit in Problem-3, reading of $V_{\text{A}}$ is done with the circuit shown next.

5. What is the current that flows through $R_5$ into the non-inverting input? **[0.5p]** Express the output voltage in terms of the bridge voltage that is $V_{\text{out}}(V_{\text{A}})$! **[1p]**